

A Unified Integral for Wave Optics

Interference and Diffraction as Two Faces of One Equation

research note

What this note is about. Interference and diffraction are not two phenomena with two separate formulas. They are **one and the same integral**, degenerating differently as the parameters (a, d, N) take different limits: the slit-width limit $a \rightarrow 0$ is interference, the single-aperture case $N = 1$ is diffraction, and the general case sits in between. The whole note rests on a single physical input (point sources emit waves, phase advances with distance, contributions add up); everything else is computing that integral. Every “textbook formula” appears as **the product of some approximation**, never as a premise.

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1 Starting point: only three things granted

The whole model needs no electromagnetic field, no energy, no complicated physical ontology. It grants only three statements — and each is first justified by **why**, then written as a formula.

(1) Every point on a wavefront is a fresh little source. Intuition: as a wave advances, every point of the wavefront “relays” the wave forward by emitting its own little spherical wavelet; the next wavefront is the envelope those wavelets add up to. So **a slit can be treated as packed with infinitely many point sources** — exactly what licenses “slicing the slit into points and integrating” later.

(2) Phase advances with distance: one full turn per wavelength. Intuition: as the wave travels forward, every **one wavelength** λ the waveform repeats — the phase turns through a full 2π . Since “travel λ , turn 2π ,” the phase angle is proportional to the distance travelled. Travel r and it turns

$$\phi = \frac{2\pi}{\lambda} r.$$

The constant $k \equiv \frac{2\pi}{\lambda}$ (phase turned per unit distance) is used throughout.

(3) Waves reaching one point simply add. Intuition: two waves arriving at the same point add their displacements — in phase (crest on crest) they reinforce, out of phase (crest on trough) they cancel. Writing each wave’s phase as a complex number $e^{i\phi}$ (one turn = 2π , consistent with (2)), “adding” becomes complex summation, and the total disturbance at a screen point is the sum over all sources:

$$U = \sum (\text{sources}) e^{i\phi} \xrightarrow{\text{continuum}} \int e^{i\phi} (\dots).$$

That is all three. The rest is mathematics. This is the sole physical input; no new assumption is introduced afterwards. “Interference” and “diffraction” are two faces of this one integral under different parameters — that is what the note sets out to make clear.

2 The master equation

Coordinates: the aperture sits on the y -axis, the screen at $x = L$, the observation point $P(L, y_P)$. The distance from an aperture point $(0, y)$ to P is taken honestly by Pythagoras, **no approximation**:

$$r(y) = \sqrt{L^2 + (y_P - y)^2}.$$

The aperture: N slits, each of width a , with a solid wall of width d between adjacent slits (so adjacent slit **centres** are $d + a$ apart), symmetric about the origin, with one slit centred at 0. The centre of slit j :

$$c_j = j(d + a), \quad j = 0, \pm 1, \pm 2, \dots$$

The total complex amplitude at P is then “integrate wherever it is transparent”:

$$U(y_P) = \sum_j \int_{c_j - a/2}^{c_j + a/2} e^{ikr(y)} dy, \quad k \equiv \frac{2\pi}{\lambda}$$

All three parameters (a, d, N) live here. The intensity is $I = |U|^2$, and **the darkest point is where** $U = 0$. (The per-unit-length amplitude density is a global constant that does not affect the zeros; absorb it.) This single equation is the bedrock of the note; every result below is just this integral, evaluated for different parameters.

Notation note. Here d is the **gap** between slits, so the centre-to-centre spacing (period) is $d + a$. Textbooks often use the symbol d for the **period** itself; the two differ by a . Below I write the period as $p \equiv d + a$; mind the conversion.

3 Pushing the integral through: series expansion

The integral with the variable buried in a square root is awkward. First shift each slit's integration variable to its own centre: $y = c_j + u$, $u \in [-\frac{a}{2}, \frac{a}{2}]$. Define the centre-to- P distance (a constant, independent of u)

$$R_j = \sqrt{L^2 + (y_P - c_j)^2},$$

and a pure number

$$s_j = \frac{y_P - c_j}{R_j} \in (-1, 1),$$

geometrically the “slope of the direction from the slit centre toward P .” (It will turn out to be the textbook $\sin \theta$, but treat it as an algebraic quantity for now.)

Pull R_j out of $r_j(u) = \sqrt{R_j^2 - 2(y_P - c_j)u + u^2}$ and expand with $\sqrt{1-x} = 1 - \frac{1}{2}x - \frac{1}{8}x^2 - \dots$ to get an **exact** power series in u :

$$r_j(u) = R_j \underbrace{-s_j u}_{u^1} + \underbrace{\frac{1-s_j^2}{2R_j} u^2}_{u^2} + \underbrace{\frac{s_j(1-s_j^2)}{2R_j^2} u^3}_{u^3} + \dots$$

This is the technical core of the note: **the distance from each interior point of the slit to P is written as “slit-centre distance R_j ” plus “corrections in the offset u of every order.”** The first term R_j governs what happens *between* slits; the u -terms govern what happens *inside* one slit. A single series splits the two into different powers — which is precisely why they are one and the same thing.

4 Structure of the exact solution

Split the phase into a “slit-centre part” (pulled out of the integral) and an “interior part”:

$$U(y_P) = \sum_j e^{ik R_j} \int_{-a/2}^{a/2} \exp\left(ik \left[-s_j u + \frac{1-s_j^2}{2R_j} u^2 + \dots\right]\right) du.$$

Call the inner integral $W_j(a)$. Expand its exponential as a power series and integrate term by term. On a symmetric interval, odd powers vanish and even powers do not:

$$\int_{-a/2}^{a/2} u^p du = \frac{2}{p+1} \left(\frac{a}{2}\right)^{p+1} \quad (p \text{ even}), \quad 0 \quad (p \text{ odd}).$$

So each slit's inner factor is a series in odd powers of a only:

$$W_j(a) = a + c_2 a^3 + c_4 a^5 + \dots = a[1 + O(a^2)],$$

with coefficients set by s_j, R_j . Together:

$$U(y_P) = \sum_j e^{ik R_j} W_j(a), \quad W_j(a) = a[1 + O(a^2)]$$

Reading this:

- $\sum_j e^{ik R_j}$ — depends only on **slit-centre positions**. Each of the N centres contributes a unit-length phasor $e^{ik R_j}$; they add. Parameters d, N live here.
- $W_j(a)$ — depends only on **a single slit's width**. It is a series in a , tending to a as $a \rightarrow 0$ (the same constant for every slit). Parameter a lives here.

- The two **multiply** inside each term. But W_j carries the index j : different slits have different W_j (different s_j, R_j), so at this stage one **cannot** pull W out of the sum into a clean “between-slit $\times$ within-slit” product. When it *can* be pulled out, i.e. when the textbook product formula holds, is something to be computed later, not assumed now.

Up to here everything is exact. What follows is just evaluating it in different parameter regimes.

5 The collapse limit $a \rightarrow 0$: two point sources, an exact hyperbola

Take $a \rightarrow 0$: each slit’s inner factor $W_j(a) \rightarrow a$, **the same constant** for all slits, pulled out of the sum (and dropped — it does not affect zeros). What remains is a pure sum over slit centres:

$$U(y_P) = \sum_j e^{ikR_j}, \quad R_j = \sqrt{L^2 + (y_P - c_j)^2}.$$

(After $a \rightarrow 0$ the period $p = d + a \rightarrow d$, so the point-source spacing is just d .) Take the simplest case $N = 2$. Two point sources, spacing denoted D , at $\pm \frac{D}{2}$:

$$r_1 = \sqrt{L^2 + (y_P - \frac{D}{2})^2}, \quad r_2 = \sqrt{L^2 + (y_P + \frac{D}{2})^2}.$$

Set $U = 0$ **directly, with no approximation**:

$$e^{ikr_1} + e^{ikr_2} = 0 \iff e^{ik(r_2 - r_1)} = -1 \iff k(r_2 - r_1) = (2m + 1)\pi.$$

With $k = \frac{2\pi}{\lambda}$:

$$r_2 - r_1 = \left(m + \frac{1}{2}\right)\lambda \text{ (dark, exact);} \quad r_2 - r_1 = m\lambda \text{ (bright)}$$

This is the very conclusion that the “exact geometric derivation” was after, now falling straight out of the integral — no separate construction needed. The locus of points with a constant difference of distances to two fixed points is a **hyperbola** with the two sources as foci: dark fringes are the hyperbolae with difference $(m + \frac{1}{2})\lambda$, bright ones with $m\lambda$. The screen $x = L$ cuts them at the exact positions. Solving for the bright-fringe coordinate:

$$y_P = \pm \frac{m\lambda}{2} \sqrt{1 + \frac{4L^2}{D^2 - m^2\lambda^2}} \quad (\text{no approximation}).$$

5.1 Where the textbook $d \sin \theta$ comes from

The exact path difference can be written as a clean fraction (difference of squares), no approximation:

$$r_2 - r_1 = \frac{r_2^2 - r_1^2}{r_2 + r_1} = \frac{2y_P D}{r_2 + r_1}.$$

Only when the denominator $r_2 + r_1$ is crudely replaced by the constant $2L$ (i.e. P is taken to be roughly equidistant from both sources) does one get

$$r_2 - r_1 \approx \frac{2y_P D}{2L} = D \cdot \frac{y_P}{L} \approx D \sin \theta,$$

hence bright at $D \sin \theta = m\lambda$. **The textbook formula replaces the hyperbola by its asymptote** — approximating the denominator by a constant is the entire cost.

6 General N : no closed form exactly, the grating condition after approximation

The interference limit for general N :

$$U(y_P) = \sum_{j=0}^{N-1} e^{ikR_j}, \quad R_j = \sqrt{L^2 + (y_P - jD)^2}.$$

The catch: the phases kR_j are **not in arithmetic progression** (R_j is a square root in j , nonlinear). Not arithmetic $\Rightarrow$ not geometric series $\Rightarrow$ setting $U = 0$ exactly has **no clean closed form**. $N = 2$ was solvable because the ratio of the two terms carries a single phase; $N \geq 3$ is not.

Those neat grating formulas all rest on **forcing the phase to be linear**. Linearize R_j about the slit centres ($R_j \approx R_0 - s(jD)$, with $s = \sin \theta$ taken common to all slits); the phases become arithmetic and the sum collapses to a geometric series:

$$U \approx e^{ikR_0} \sum_{j=0}^{N-1} e^{-ij\varphi} = e^{ikR_0} \frac{1 - e^{-iN\varphi}}{1 - e^{-i\varphi}}, \quad \varphi \equiv kD \sin \theta = \frac{2\pi D \sin \theta}{\lambda}.$$

Taking the modulus:

$$I = |U|^2 \propto \left(\frac{\sin(N\varphi/2)}{\sin(\varphi/2)} \right)^2.$$

- **Principal maxima** (brightest): $\varphi = 2\pi m$, i.e. $D \sin \theta = m\lambda$.
- **Dark fringes** ($U = 0$): $\frac{N\varphi}{2} = m'\pi$ with m' not a multiple of N , i.e. $ND \sin \theta = m'\lambda$. There are $N - 1$ dark fringes between adjacent principal maxima; larger N sharpens the maxima.

These “standard grating conditions” **appear only after linearization**. Exactly (phases nonlinear, not arithmetic), $U = 0$ for $N \geq 3$ has no closed form, the darkest point is not an absolute zero, and its position drifts with L . This is the same story as single-slit diffraction (next section): **linearize the phase in position and you get the clean textbook formula; refuse to, and the zeros are neither clean nor exactly zero.**

7 The single-slit limit $N = 1$: compute the zero directly, no pairing

The other end: $N = 1$ (or $d \rightarrow 0$, merging all slits into one wide slit) leaves a single term, back to a single-slit integral:

$$U(y_P) = \int_{-a/2}^{a/2} e^{ikr(y)} dy, \quad r(y) = \sqrt{L^2 + (y_P - y)^2}.$$

Here we **do not tell the “pair the top edge with the midpoint, cancel two by two” story** — that narrative is reverse-engineered from the answer and quietly assumes the phase is linear across the slit. To ask “is the darkest point actually 0,” just compute this integral.

7.1 Keep only the linear term: the textbook $a \sin \theta = m\lambda$

Expand the phase only to first order in u ($r(y) \approx R_0 - s u$, $s = \sin \theta$):

$$U \approx e^{ikR_0} \int_{-a/2}^{a/2} e^{-iks u} du = e^{ikR_0} a \frac{\sin(\frac{ksa}{2})}{\frac{ksa}{2}}.$$

Zeros at $\frac{ksa}{2} = m\pi$, i.e.

$$a \sin \theta = m\lambda \quad (m = \pm 1, \pm 2, \dots)$$

This is the textbook single-slit dark condition. **Note where it comes from: keeping only the linear term.** So $a \sin \theta = m\lambda$ is itself an approximation, not the exact edge-to-edge path difference.

7.2 Why this is “darkest” (from integral geometry, not pairing)

Slice the slit into many pieces; each contributes a unit-length little arrow whose direction is set by its phase. When the phase is **linear** in u , the arrows turn at a **constant angular rate** and, laid head to tail, trace a **circular arc**; the resultant is the **chord** of that arc. When the total phase across the slit, $ksa = \frac{2\pi}{\lambda}a \sin \theta$, is an integer multiple of 2π , the arrows wind through whole **closed loops**, head meets tail, and the chord length is 0 $\Rightarrow$ full dark. That is the geometric meaning of $a \sin \theta = m\lambda$ — **no pairing story needed; it is the fact that continuous arrows close into a loop that pins the zero.**

7.3 Without approximation: the darkest point is not zero, and it drifts with L

Now add the u^2 and higher terms back. The phase is no longer linear in $u \Rightarrow$ the little arrows turn at a **non-uniform rate** $\Rightarrow$ what they trace is not a true circular arc $\Rightarrow$ even when the total phase nominally “winds through a full turn,” head does **not** meet tail, and the chord is nonzero. Hence:

Under the exact integral, the single-slit “darkest point” is **generally no longer exactly zero**, and its position drifts with the parameters. That quadratic coefficient $\frac{1-s^2}{2R} = \frac{\cos^2 \theta}{2R}$ carries $R \sim L$ — so “how dark” is a **function of the observation distance L** : larger L makes the quadratic term smaller, the arrow-arc closer to a true circle, the minimum closer to true black; smaller L leaves a brighter residue in the dark. The textbook “dark-fringe intensity is exactly 0” is what you get by forcing the arrows into a perfect circle (keeping only the linear term).

This answers the original counterintuitive point head-on: **the clean $a \sin \theta = m\lambda$ comes from the linear term; “exactly zero” is a by-product of that approximation; add the higher terms and the zero is filled in and pushed off.** Single-slit diffraction never had a true absolute-dark point — only a minimum whose depth varies with L .

8 The three parameters together

One integral $U(y_P) = \sum_j e^{ikR_j} W_j(a)$; each parameter governs its own piece, and different limits are different “phenomena”:

Parameter limit	U degenerates to	Usual name
$a \rightarrow 0, N = 2$	$e^{ikR_1} + e^{ikR_2}$	two-source interference (exact hyperbola)
$a \rightarrow 0, N$ general	$\sum_j e^{ikR_j}$	N -slit interference / grating
$a \rightarrow 0, N$ large	same, maxima sharpen	high-resolution grating
$N = 1$ (or $d \rightarrow 0$)	$\int_{-a/2}^{a/2} e^{ikr} dy$	single-slit diffraction
a, d, N all finite	$\sum_j e^{ikR_j} W_j(a)$	general case (both effects)

The two ends are two degeneration directions of the same U :

- $a \rightarrow 0$: the inner factor W_j becomes the same constant a for all slits, pulled out of the sum, so slit width drops out entirely, leaving the pure sum over slit centres $\sum_j e^{ikR_j}$ — this is **interference**.
- $N = 1$: there is nothing to sum over, leaving only the integral inside one slit $W_0(a)$ — this is **diffraction**.

So “the standalone geometric derivation of interference can be dropped” holds: the hyperbola is the product of $a \rightarrow 0, N = 2$ set to $U = 0$, and the geometric construction is merely one visualization of it, already subsumed by this framework.

9 The general case: product structure and missing orders (both approximation artifacts)

Back to finite a, d, N . When can U be written as a clean “between-slit factor $\times$ within-slit factor”? Answer: **after linearizing the phase in position**. Then all $s_j \approx s$ (one common $\sin \theta$), the inner factor W_j decouples from j into a common factor $W(a)$, and the outer sum is geometric:

$$U \approx \underbrace{W(a)}_{\text{single slit, } a} \times \underbrace{\sum_j e^{-ij\varphi}}_{N \text{ slits, } p=d+a, N}, \quad \varphi = \frac{2\pi p \sin \theta}{\lambda}.$$

Here $|W(a)|$ is the single-slit $\left| \frac{\sin(\pi a \sin \theta / \lambda)}{\pi a \sin \theta / \lambda} \right|$ and the multi-slit modulus is $\left| \frac{\sin(N\varphi/2)}{\sin(\varphi/2)} \right|$.

Missing orders are where the zeros of this product coincide: multi-slit principal maxima at $p \sin \theta = m\lambda$, single-slit zeros at $a \sin \theta = q\lambda$. When they land at the same angle, a principal maximum that “should” be there is forced dark by the single-slit zero, and that order goes missing. The condition:

$$\frac{p}{a} = \frac{m}{q} \text{ an integer } n \implies \text{orders } n, 2n, 3n, \dots \text{ missing} \quad (p = d + a).$$

(With the textbook symbol $p \rightarrow d$, this is the familiar $d/a = n$.)

The point: **the product structure and missing orders are not built into the master equation; they are consequences of linearization**. In the master equation W_j still carries its index j and is entangled with the outer sum; only after approximation do they decouple into a tidy product.

10 Beyond the one physical input: wavelength and white light

Everything above fixed a single wavelength λ . Real white light is a mix of wavelengths with a power spectrum $f(\lambda)$. Since all of U depends on λ (through $k = 2\pi/\lambda$ in the phase), the total screen intensity is the weighted superposition of the per-wavelength patterns:

$$I_{\text{total}}(y_P) = \int_0^\infty |U(y_P; \lambda)|^2 f(\lambda) d\lambda.$$

The maxima / zeros of every wavelength shift with λ (since $\sin \theta \propto \lambda$):

- Centre ($m = 0$): all wavelengths undeflected and coincident, still white.
- Sides: longer wavelength deflects more, red outside, violet inside; each order spreads into a coloured band, wider and more prone to overlap with neighbours at higher order.

That is the whole origin of white-light fringes fanning into a rainbow — no new physics, just letting λ go from a constant to an integration variable.

11 Summary

1. **One equation:** $U(y_P) = \sum_j \int_{c_j-a/2}^{c_j+a/2} e^{ikr} dy$. Point sources, phase, superposition — three assumptions, the rest is mathematics.
2. **One expansion:** writing the interior-point distance as “slit-centre distance + corrections in the offset” gives $U = \sum_j e^{ikR_j} W_j(a)$ — the between-slit effect (e^{ikR_j}) and the within-slit effect (W_j) fall automatically into different powers.
3. **Two limits:** $a \rightarrow 0$ yields interference ($N = 2$ solves exactly to a hyperbola, and $d \sin \theta = m\lambda$ is its asymptote); $N = 1$ yields diffraction ($a \sin \theta = m\lambda$ comes from keeping only the linear term).

4. **One honest conclusion:** every “clean formula” is the product of **linearizing** the phase in position. Without that, neither $N \geq 3$ interference nor single-slit diffraction has a closed-form zero; the darkest point is **not an absolute zero** and drifts with L . The textbook “dark-fringe intensity is zero” is the illusion of forcing the continuous phasors into a perfect circle.